

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Fundamentals of Data Science **COURSE CODE:** DSA04

COURSE OUTCOME COVERED

CO1: Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)

QUESTIONS: 05 **Total Marks:** 100

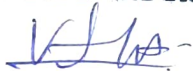
Annexure A

CONCEPT MAPPING

Code of Conduct:

I Vivekha B C 192424130 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Concept Mapping

- 1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.**
- 2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.**
- 3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.**
- 4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.**
- 5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.**

RUBRICS FOR CALCULATION

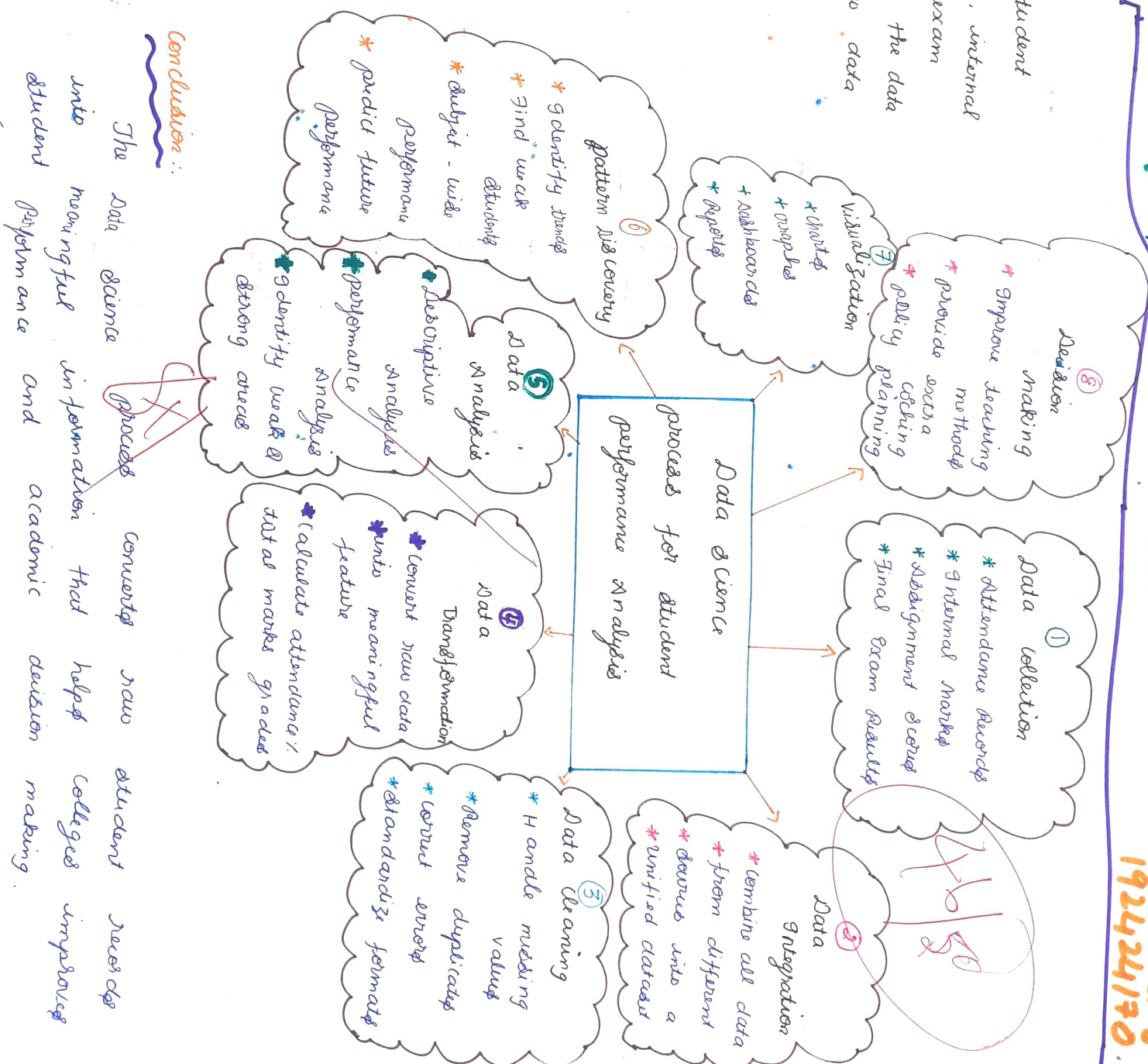
Organizational	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	5% 10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores and final exam results. Draw a concept map showing the data science process from data collection to data transformation.

Definition:

Data science is the process of collecting, cleaning, processing, analyzing and transforming data into useful information for decision-making.

Sm that it include Data collection, Data integration, Data cleaning, Data preprocessing, Data Transformation, Data Analysis, pattern discovery, Visualization and Decision making like improve teaching arrange classes and performance.

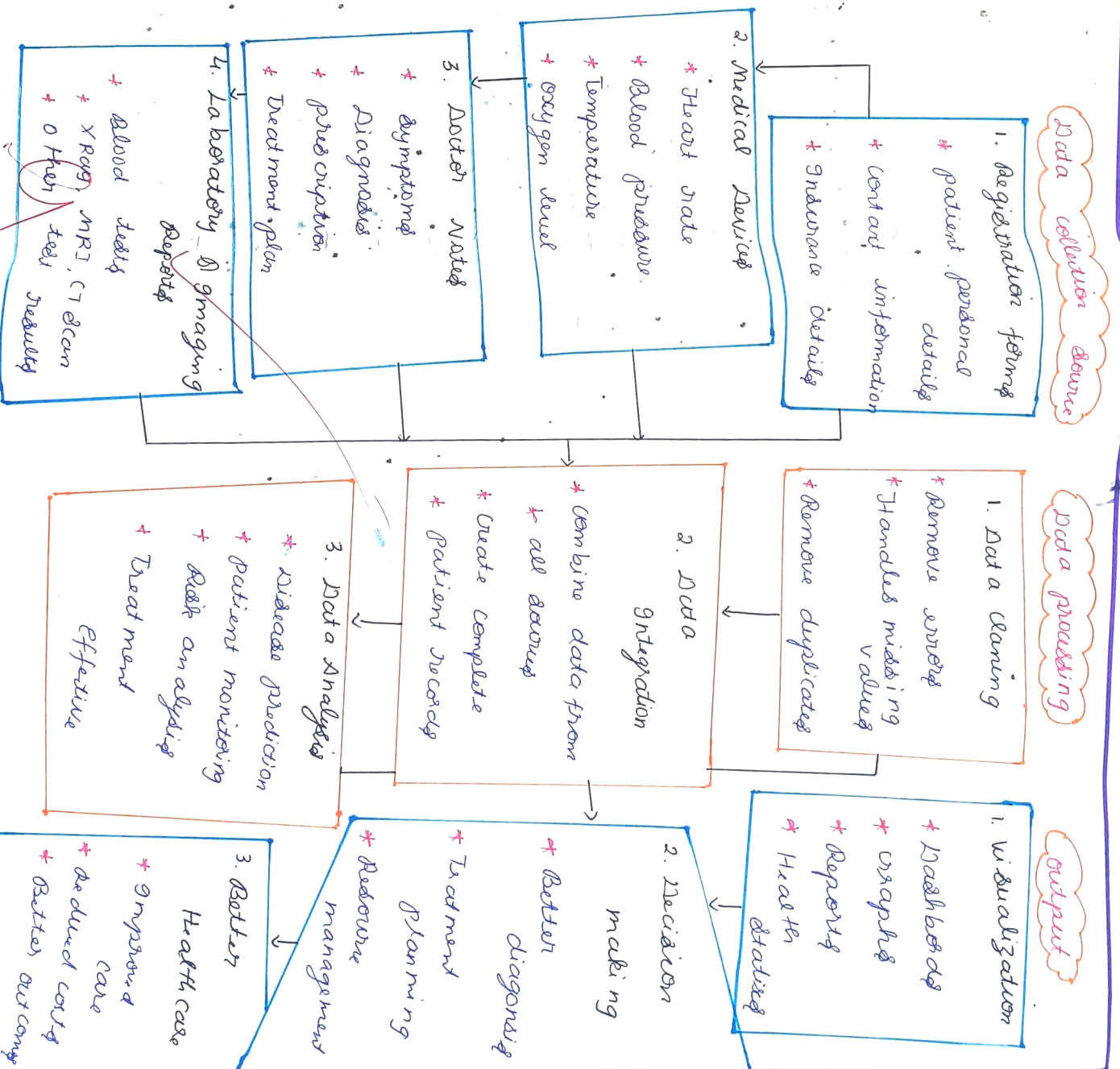


2. A Hospital collects patient details through registration forms, medical devices, and doctor notes. Construct a concept map showing different data collection sources and how they are used for analysis.

Definition:

Healthcare Data Science collects patient information from multiple source and analysed it to improve diagnosis, treatment and healthcare management.

In this it include the patient data sources, Registration Notes, forms, Devices and Doctor Reports also it is work with data collection, Data cleaning, Data integration, Data analysis and visualization.

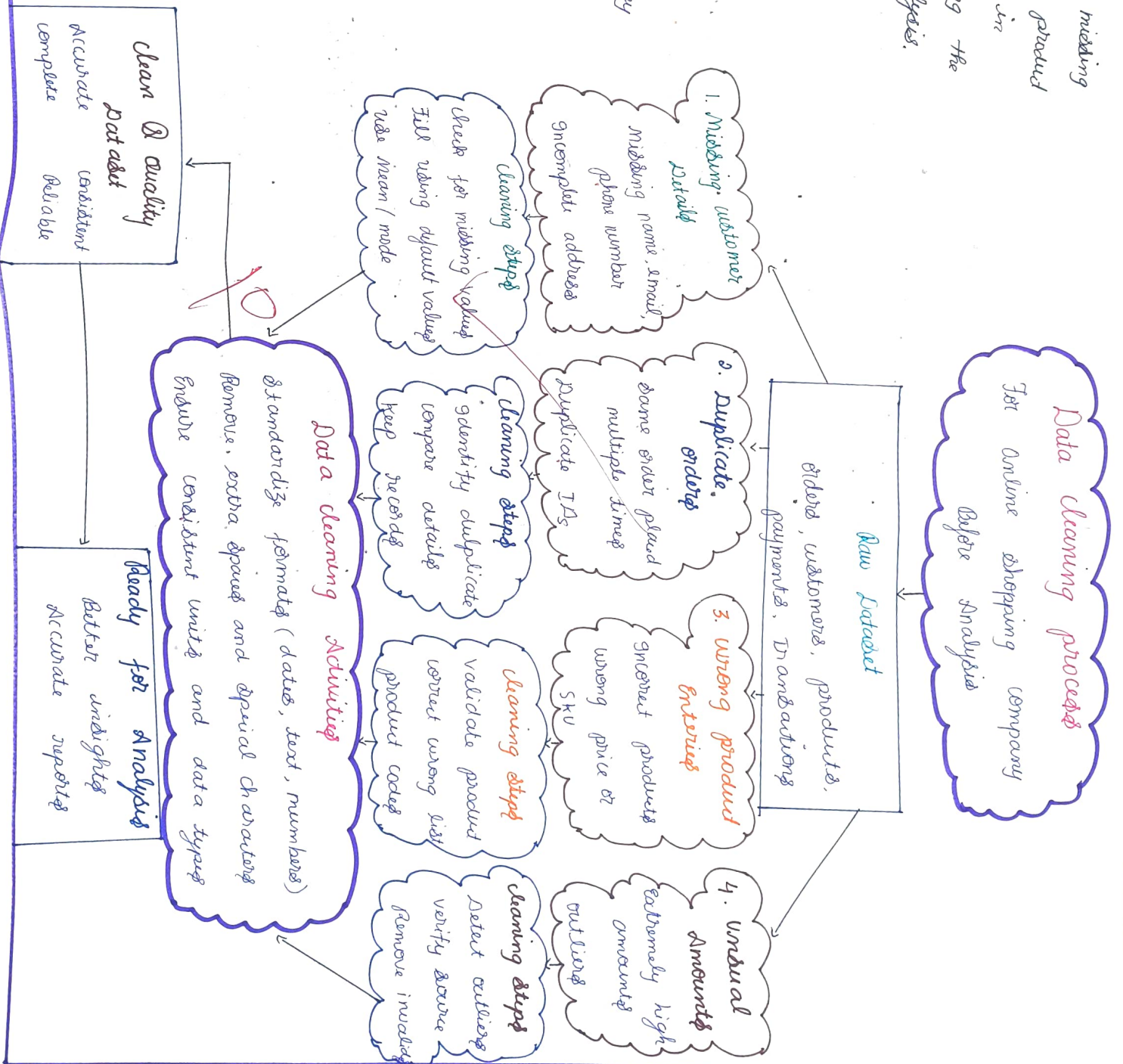


3) An Online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.

Definition

Data cleaning is an important step in the data science process that improves the quality of data before analysis. In an online shopping company the dataset may contain missing customer details, duplicate orders, incorrect product.

A clean dataset helps the company perform accurate analysis, improve customer insights, and make better business decisions. Unusual purchase amount (outliers) are detected and checked to determine whether they are valid or errors. To make the correct decisions.

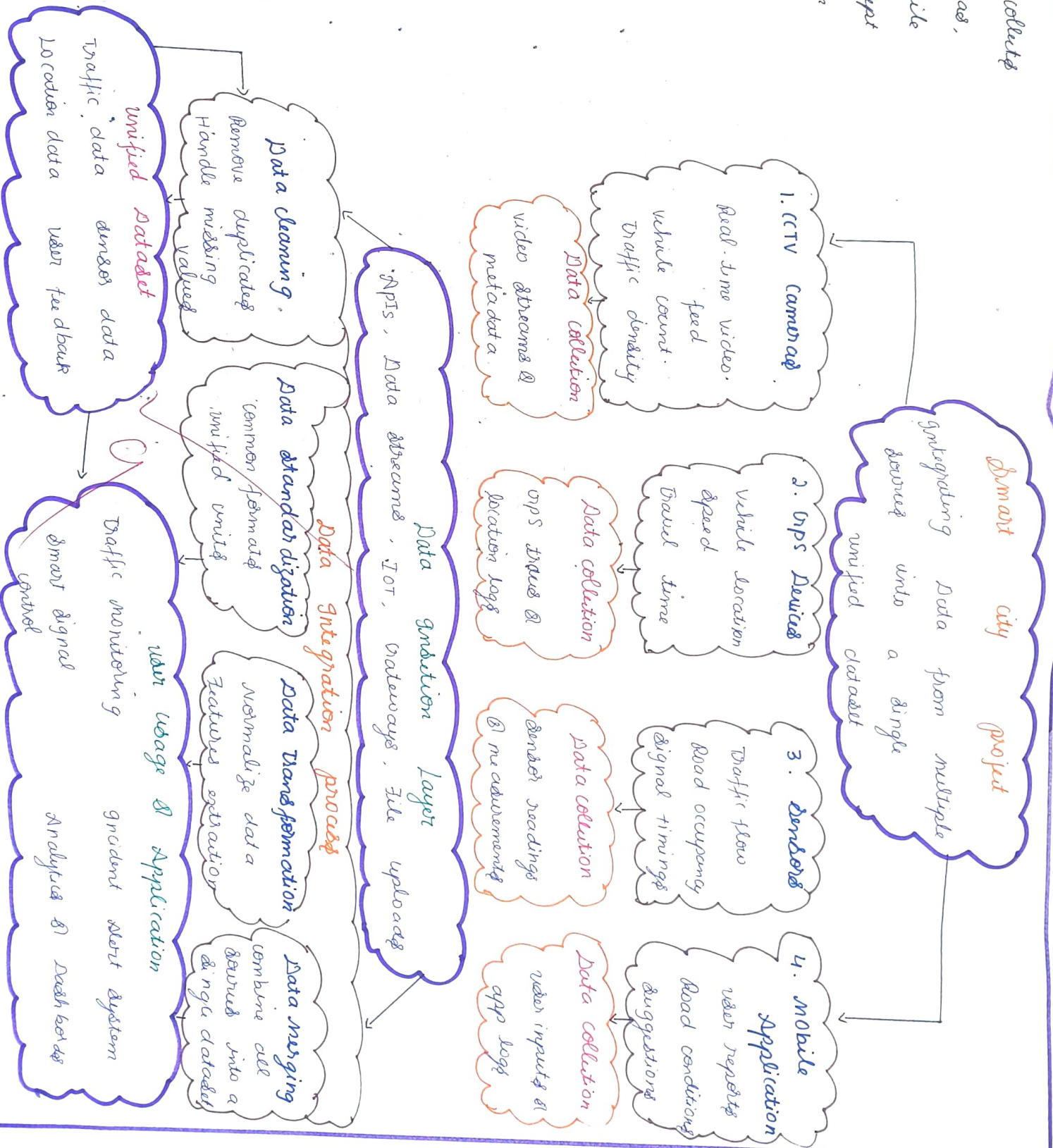


4) A smart city project collects traffic data from CCTV cameras, GPS devices, sensors and mobile applications. prepare a concept map showing how data from multiple sources can be integrated into a single dataset.

Definition:

A smart city project collects traffic data from multiple sources such as CCTV cameras, GPS devices, sensors and mobile applications.

The collected data is cleaned, standardized and integrated into a single unified dataset for the efficient smart city transportation.



5) A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

Definition:

Data transformation preparing customer data for accurate loan approval prediction. Numerical attributes such as income, age, loan amount & credit score are normalized. These are improved data quality in create model accuracy and help machine learning.

